Performance Task: Feedback Control System Prototype

Course: Feedback and Control Systems | BS Computer Engineering, Third Year | Duration: 2–3 weeks

1. Performance task description

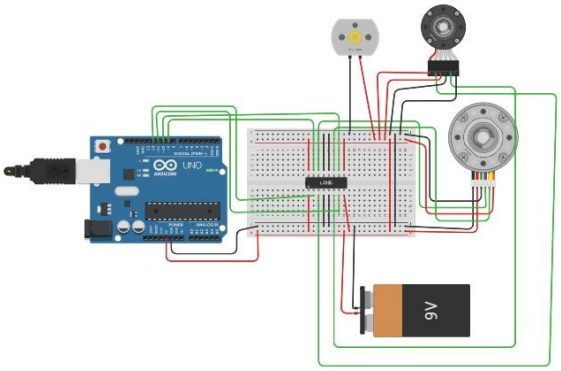
Students, working in groups of 3–4, will design, simulate, construct, and test a simple feedback control system for a selected engineering application. They will identify the plant, input, output, sensor, and actuator; develop a basic mathematical model; and obtain or estimate the plant transfer function when feasible.

Students will implement and demonstrate both open-loop and closed-loop operation, using a simple controller such as on-off, proportional (P), or proportional-integral (PI) control. They will observe and compare the system responses and explain how feedback affects control accuracy and system performance.

The project will culminate in a functional prototype, a short technical report, and a live demonstration. The emphasis is on demonstrating fundamental control concepts rather than developing a sophisticated or publication-level system.

2. Suggested project topics

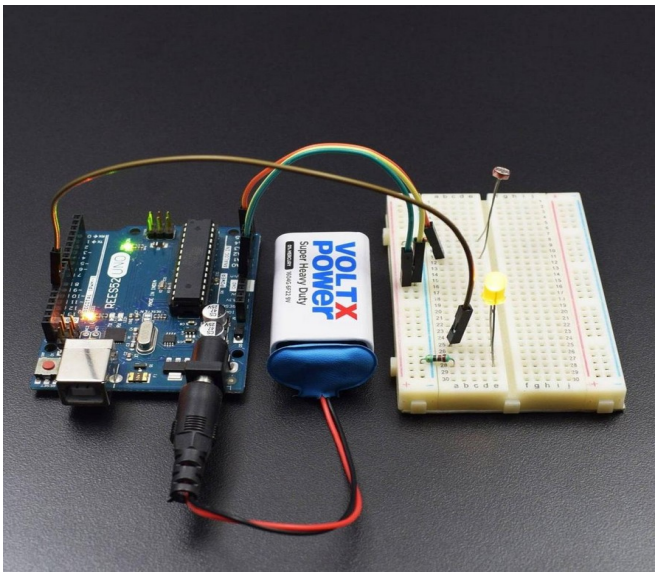
These topics are achievable with commonly available laboratory components.



A. DC Motor Speed Control — Recommended

Maintain the speed of a small DC motor using an encoder or speed sensor and a micro-controller. Compare fixed PWM operation with feedback-based speed regulation.

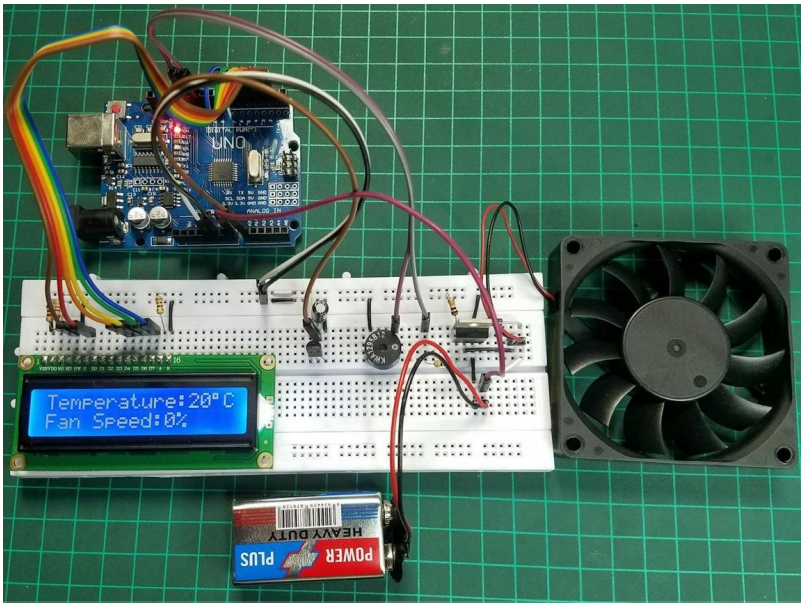
Controller: P or PI | Difficulty: Moderate



B. Automatic Light Intensity Control

Maintain a target light intensity using an LDR or light sensor and an LED controlled through PWM.

Controller: On-off or P | Difficulty: Easy



C. Temperature Control System

Maintain a desired temperature using a temperature sensor and a low-voltage fan or an appropriately protected low-voltage heating element.

Controller: On-off | Difficulty: Easy to moderate



D. Automatic Water-Level Control

Maintain the water level in a small tank using a level sensor and a low-voltage DC pump.

Controller: On-off | Difficulty: Easy to moderate

For the shortest implementation period, automatic light intensity control or temperature control is the easiest to complete. DC motor speed control is preferable if students have already worked with motor drivers and encoders.

3. Required procedure

Students must complete the following six activities.

Step	Activity and instructions	Required output
1	Select a topic, identify the control problem, and draw a simple block diagram showing the reference input, controller, plant, sensor, and feedback path.	Block diagram and component list
2	Develop a basic mathematical model and obtain the transfer function using governing equations or an appropriate simplified model. Explain the assumptions.	Equations and transfer function
3	Simulate or predict the plant's response to a step input using an approved software tool.	One open-loop response plot
4	Build the prototype and implement a simple feedback controller. Test open-loop and closed-loop operation.	Functional prototype and source code
5	Apply at least two reference settings and compare the measured outputs in both configurations. Record the results and explain whether feedback improves performance.	Data table and response comparison
6	Submit a short report and demonstrate the working prototype to the instructor.	Short report and live demonstration

4. Required open-loop and closed-loop comparison

Students should record at least three measurements per configuration where practical.

Test condition	Open-loop	Closed-loop
Reference or target setting	Record	Record
Measured output	Record	Record
Absolute steady-state error, if applicable	Calculate	Calculate
Response to a change in target	Observe	Observe
Response to a disturbance, if safe and applicable	Observe	Observe

Students must explain which configuration tracks the target more accurately and whether the improvement is consistent with their model and simulation. Formal calculation of rise time, settling time, and overshoot may be optional for simpler on-off systems.

5. Short technical report format

Limit the report to approximately 4–6 pages, excluding code and appendices.

1. Introduction and objectives — Describe the application, control problem, and objectives.
2. System design and mathematical model — Include the block diagram, components, governing equation, and transfer function or estimated model.
3. Prototype development and procedure — Briefly explain the circuit, controller, and testing procedure.
4. Results and discussion — Include simulation results, experimental measurements, and the open-loop versus closed-loop comparison.
5. Conclusion — State whether the objectives were achieved and identify one limitation or possible improvement.

Append only the necessary wiring diagram and source code. A separate literature review, lengthy theoretical chapter, and extensive research documentation are not required.

6. Assessment rubric

Total: 100 points

Criterion	Assessment focus	Points
System design and block diagram	Correct identification of the plant, sensor, actuator, input, output, and feedback path	10
Mathematical model and transfer function	Reasonable equations, correct derivation or estimation, and stated assumptions	15
Simulation and response analysis	Appropriate open-loop response and basic interpretation	10
Functional prototype	Working hardware, sensor integration, controller, and safe operation	30
Testing and performance comparison	Actual measurements and a reasonable comparison of open-loop and closed-loop performance	15
Short technical report	Clear, complete, and technically consistent documentation	10
Demonstration and explanation	Successful demonstration and ability to explain how the feedback works	10
Total		100

The prototype receives the highest weight because the principal deliverable is a functioning system supported by basic control theory.

7. Two-to-three-week schedule

Week 1: Design and modeling

a. Select the project topic and components.
b. Draw the block diagram.
c. Develop the basic model and transfer function.
d. Complete the initial open-loop simulation.

Week 2: Construction and testing

a. Assemble the prototype.
b. Implement the feedback controller.
c. Test open-loop and closed-loop operation.
d. Record measurements and correct basic hardware or software problems.

Week 3: Final evaluation and submission

A. Complete performance comparisons.
B. Submit the 4–6 page report.
C. Demonstrate the prototype and answer instructor questions.

8. Final submission requirements

Each group must submit only the following:

a) Functional feedback control prototype.
b) System block diagram and circuit or wiring diagram.

c) Basic mathematical model and transfer function derivation or estimation.
d) One open-loop simulation plot.
e) Experimental comparison table for open-loop and closed-loop operation.
f) Short technical report of approximately 4–6 pages.
g) Source code and live demonstration.

Recommended final arrangement: Use a three-week maximum, groups of 3–4 students, one approved topic per group, a short report, and a live demonstration (recorded video demonstration). Require mathematical modeling and transfer function analysis, but keep them proportional to the complexity of the chosen plant. This preserves the essential learning outcomes of Feedback and Control Systems without turning the performance task into a full research project.